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EDITED BY

Rowena Christiansen,
The University of Melbourne, Australia

REVIEWED BY

Ilaria Cinelli,
Aerospace Medical Association, United States
Priya Patel,
Saurashtra University, India

*CORRESPONDENCE

Alina Kunitskaya,
✉ alinakunitskaya@gmail.com

[†]These authors have contributed equally to this work and share first authorship

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The database of peptide medications for human deep-space missions demonstrates the potential for on-demand "Astropharmacy"

Rebecca C. Blum^{1,2†}, Dorra Omrani^{1†}, Alina Kunitskaya^{3,4*}, Jessica E. Snyder^{2,4}, Cynthia Bui¹, Philip M. Williams⁵, David J. Loftus⁶ and Lynn J. Rothschild^{1,4,7}

¹Department of Molecular Biology, Cell Biology and Biochemistry, Brown University, Providence, RI, United States, ²Blue Marble Space Institute of Science, Seattle, WA, United States, ³NASA Postdoctoral Program/Oak Ridge Associated Universities, Oak Ridge, TN, United States, ⁴NASA Ames Research Center, Planetary Systems Branch, Moffett Field, CA, United States, ⁵School of Pharmacy, University of Nottingham, Nottingham, United Kingdom, ⁶NASA Ames Research Center, Space Biosciences Research Branch, Moffett Field, CA, United States, ⁷NASA Ames Research Center, Space Science and Astrobiology Division, Moffett Field, CA, United States

With the advent of long-duration lunar and Martian space missions, health risks will increase significantly as astronauts will be exposed to prolonged periods of reduced gravity, elevated radiation levels, celestial dust, and isolated environments that may interfere with psychological health and sleep. Astronauts will increasingly rely on pharmaceutical intervention to mitigate these health risks, particularly drugs targeted to treat spaceflight-induced medical conditions such as bone loss, cytopenias, and other degenerative conditions secondary to radiation exposure and immune system dysfunction. Many of these are biologic drugs, and peptide and protein pharmaceuticals are particularly unstable, with limited shelf life (~6 months) even with refrigeration, which is inadequate for a 2+-year mission to Mars and back. In addition to reduced drug stability in space, there are mass/volume constraints, uncertainty regarding how much—if any—of the drugs will be needed, and severely limited re-supply opportunities. The National Aeronautics and Space Administration (NASA) has identified the need to establish a drug formulary for long-duration space missions, along with a process to ensure that medications remain efficacious during flight. To address these challenges, we aim to develop an "Astropharmacy," a compact platform that uses engineered microbes to produce small doses of biologics *on demand*, either in transit or at space destinations that may be far from Earth. To assess the potential for this Astropharmacy, we have compiled a database of peptide and protein drugs that may be needed to mitigate the health risks during long-duration missions to the Moon and Mars that are well-suited for production by the Astropharmacy system. The compiled database acts as a guide for drugs that could be produced using the Astropharmacy system. Importantly, the database can also serve as a valuable resource for flight surgeons and the

medical community on Earth, particularly those providing care in resource-limited environments, along with researchers studying biologic drug development.

KEYWORDS

space medicine, peptide medications, on-demand drug manufacturing, long-duration space missions, engineered microbes

1 Introduction

During spaceflight, astronauts are exposed to prolonged reduced gravity, ionizing radiation, and an isolated, high-stress environment, which all pose significant risks to human health. The impacts of spaceflight on human health have been extensively studied and reviewed (Afshinnekoo et al., 2020; Garrett-Bakelman et al., 2019; Krittanawong et al., 2022; Kunitskaya et al., 2022). These impacts span across many body systems, including the musculoskeletal (Comfort et al., 2021; Lee Satcher et al., 2024), cardiovascular (Krittanawong et al., 2023; Shen and Frishman, 2019), immune (Akiyama et al., 2020; Crucian et al., 2018), lymphatic (Norsk et al., 2015), vestibular (Carriot et al., 2021; Clark, 2019), and nervous systems (Gupta et al., 2023; Seidler et al., 2024). In addition to physical hazards, the extreme environment of space can also result in mental and psychosocial challenges. The isolation and disruption of the circadian rhythm can lead to long-term composite stress, contributing to sleeping problems and increasing the rate of anxiety and depression in astronauts (Yin et al., 2023). All these pose risks to astronaut health and thus to mission success.

Nevertheless, astronauts have lived and worked onboard the International Space Station (ISS) in Low Earth Orbit (LEO) for the past 25 years. In addition to countermeasures (e.g., specialized diet and exercise) developed to mitigate the negative impacts of spaceflight, astronauts aboard the ISS have access to commercial over-the-counter and prescription medications for common ailments, such as headaches, sleep issues, congestion, diarrhea, and motion sickness (NASA, 2005; Wotring, 2015). Although historical medication usage aboard the ISS is poorly documented, the Dose Tracker study of six crew members reported an average of four medication uses per week per crew member, totaling 453 medication uses per crew member during the 5–6-month missions (Blue et al., 2019a). Medications are most commonly used for sleep issues, congestion, allergies, or pain, often associated with extracurricular activities or exercise (Wotring, 2015). These medications are supplied from Earth and can be easily replaced upon reaching their expiration date as part of regular resupply missions (Mehta and Bhayani, 2017). Overall, despite extensive changes to the human body in space, the health risks in LEO are considered low due to limited mission durations, effective countermeasures, a well-established medical supply, and proximity to Earth in case of emergencies.

We are entering a new era in human space exploration as the National Aeronautics and Space Administration (NASA) and other space agencies plan long-duration missions (>6 months) to the Moon and Mars. The health risks will increase significantly during these missions to deep space as astronauts will be exposed to prolonged periods of reduced gravity, elevated radiation levels, celestial dust, and isolated environments that may interfere with psychological health and sleep. Although mission durations on the ISS average ~6 months (Tomsia et al., 2024), the proposed missions

to Mars will last ~18–30 months (Figure 1). The prolonged exposure to microgravity can intensify the musculoskeletal impacts observed in LEO. For example, continued bone degeneration increases the risk of fractures, osteoporosis, and renal stone formation, conditions that are difficult to treat in flight (Sibonga et al., 2019) and can lead to medical emergencies if left untreated. The radiation dose equivalent is expected to increase from 50 to 100 mSv during a 6-month ISS mission to 300–400 mSv during 1-year sustained lunar operations, 500–600 mSv in a deep-space habitat for 1 year, and 870–1,200 mSv during a 650–920-day Mars mission (Simonsen et al., 2020). Prolonged radiation exposure affects almost all of the body's organ systems and can increase the risk of cancer through DNA damage (Guo et al., 2022). For example, radiation exposure can cause damage to the cardiovascular system through promotion of myocardial remodeling, fibrosis, and microvascular damage (Meerman et al., 2021). Additionally, short-term exposure to a high dose of radiation, such as during a solar particle event, can result in acute radiation sickness, with potentially life-threatening complications, such as neutropenia (low white blood cell count) (Guo et al., 2022). Celestial dust exposure is another challenge that was first encountered during the Apollo missions. Although Apollo missions lasted only several days, prolonged exposure to celestial dust during deep-space missions will increase the risk of ocular and pulmonary injuries (Scully et al., 2015). NASA's Human Research Program identified more than 30 human health risks associated with spaceflight (Patel et al., 2020), and most of these risks would require mitigation for lunar and Mars missions, according to NASA's Human Research Roadmap.

Far away from Earth, medical evacuation is not feasible, so astronauts increasingly rely on pharmaceutical intervention to mitigate these health risks (Blue et al., 2019a; Miller et al., 2007). However, supplying these pharmaceuticals would be challenging due to their limited shelf life, reduced drug stability in space, mass/volume constraints, and uncertainty regarding how much—if any—of the drugs would be required, all in the face of severely limited re-supply opportunities. Approximately 60% of ISS medications have a terrestrial shelf life of 36 months or less in their original packaging, and approximately 15% expire in less than 24 months (Diaz et al., 2024). However, these medications are often removed from their original packaging before spaceflight and placed into polypropylene containers to meet the mass/volume constraints. This repackaging is not protective against atmospheric factors and may contribute to accelerated drug degradation (Reichard et al., 2023). Drug degradation may be further accelerated by exposure to space radiation, variations in gravity, and vibrations (Barchetti et al., 2024; Du et al., 2011; Reichard et al., 2023; Wotring, 2016). Some drugs that have been identified by NASA as particularly susceptible to instability and bacterial growth during spaceflight include eye drops, nitroglycerin, liquid antibiotics, and insulin (NASA, 2023). Drug degradation leads to reduced potency and accumulation of impurities (drug degradation products). Commonly, drugs fail to

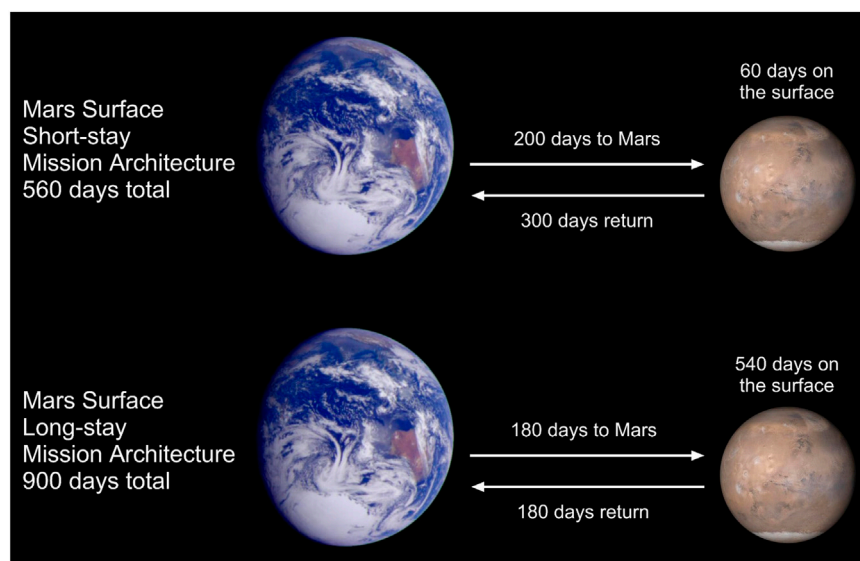


FIGURE 1

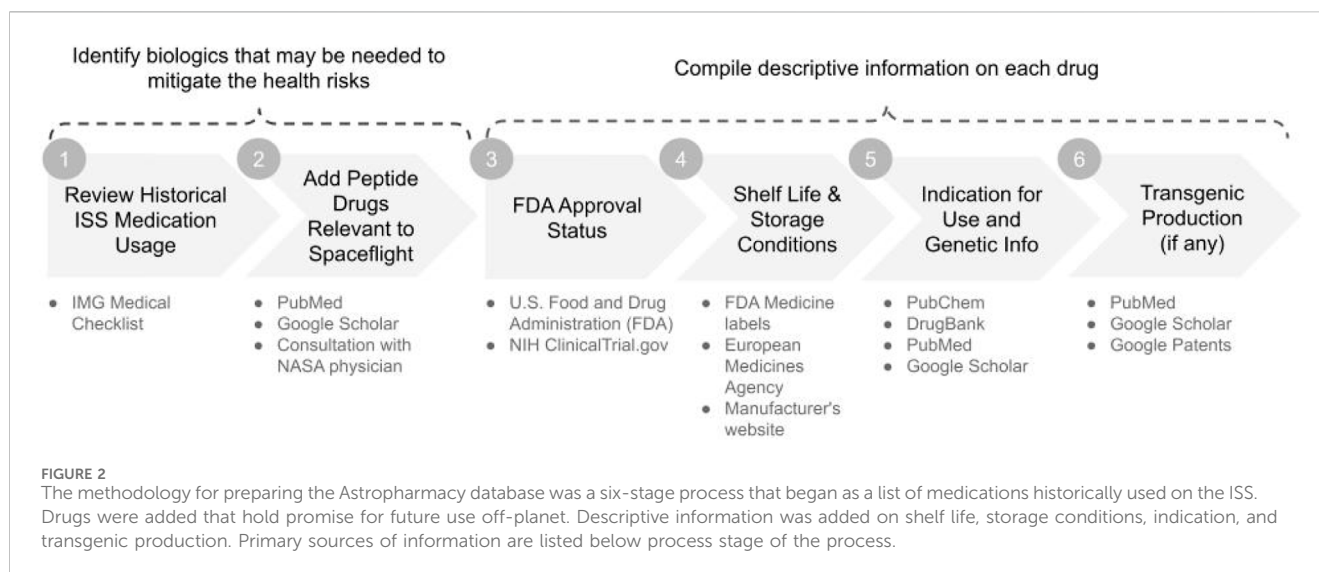
Estimated durations for short-stay (top) and long-stay (bottom) missions to the Martian surface, as described in the Human Exploration of Mars Design Reference Architecture 5.0 (Drake and Watts, 2014). Earth and Mars images were obtained from <https://images.nasa.gov/>. These are examples of deep space missions that are used as reference cases throughout the study.

meet the United States Pharmacopeia requirements for stability if the concentration of active pharmaceutical ingredients is not within 10% of the label specifications (Blue et al., 2019b). After 550 days in space, one study found that melatonin failed to meet this stability requirement, while drug degradation products were found in aspirin, ibuprofen, and amoxicillin-clavulanate (Wotring, 2016). However, studies on drug degradation in space have so far been limited by methodological challenges, such as the lack of terrestrial controls and small sample size (Reichard et al., 2023). These studies have also been completed in LEO, where the Earth's magnetic field provides some protection against radiation compared to deep-space missions. Although these concerns are mitigated on the ISS by replacing expired drugs on resupply missions, this would not be feasible on future long-duration missions, particularly to Mars. NASA's Human Research Roadmap includes the "Risk of Ineffective or Toxic Medications During Long-Duration Exploration Spaceflight" that requires mitigation for Mars missions and identifies the need to determine optimal packaging, storage, and drug formulations, along with the need for further characterization of medication stability under the spaceflight environment.

Biological drugs, pharmaceuticals derived from living organisms or their components, could be particularly well suited to treat spaceflight-induced medical conditions. In particular, peptide and protein biologics are particularly useful. On Earth, biological drugs accounted for 40% of new drugs approved by the U.S. Food and Drug Administration (FDA) in 2022 (de la Torre and Albericio, 2023), with 600 peptide drugs in preclinical studies as of 2023 (Rossino et al., 2023). The peptide drug industry is predicted to grow at a compound annual growth rate of 10% over the next 10 years (Rossino et al., 2023). Many peptide drugs on the market today are used to treat metabolic, cardiovascular, or endocrine conditions (Wang et al., 2022). Peptides have also been used for

the treatment of radiation-induced complications, such as neutropenia (Mehta et al., 2015), and investigated as radioprotective agents (Shaghghi et al., 2021). In the case of astronaut health, these peptide drugs could be used to treat bone loss, cytopenias, and other degenerative conditions secondary to radiation exposure and immune system dysfunction. Unfortunately, there are serious challenges associated with using peptide and protein biologics for long-duration flights. Primary among these is their very limited shelf life of ~6 months, which is inadequate for a 2+-year mission to Mars and back (Mazzeo and Carpenter, 2009; Ngo and Garneau-Tsodikova, 2018; Vallota-Eastman et al., 2023). Refrigeration and freezing temperatures are often required for their storage, which is challenging to provide during deep-space spaceflight due to the significant associated up-mass and volume. Additionally, spaceflight unpredictably accelerates the degradation of peptide-based drugs, indicating that such drugs have been dismissed from medical kits, despite their promising efficacy and safety. Consequently, the spaceflight community has expressed the need for in-space manufacturing of materials, including medicine, to support astronauts' health and performance in deep space (Seoane-Viano et al., 2022).

To address these challenges, several studies have focused on alternative drug formulations (e.g., the addition of photoprotective agents) and packaging to increase drug stability in space (Wani et al., 2024). We aim to investigate an alternative approach, in which required medications can be produced on-site if and when they are needed, eliminating the need for long-term storage. This approach should be capable of producing a therapeutic dose of the required medication within 24 h–48 h, use stable reagents that can be stored at room temperature, and have minimal mass, volume, power, and crew-time requirements. To achieve this, we aim to develop an "Astropharmacy," a compact platform that uses engineered microbes to produce small doses of biologics on-demand at space



destinations that may be far from Earth (Vallota-Eastman et al., 2023). Engineered microbes can be transported to deep space in the form of durable, dried spores, along with the Astropharmacy hardware. If an astronaut becomes ill, the spores can be activated to initiate production of the desired biologic. This platform would become particularly critical on missions of >6 months with limited re-supply and storage capabilities, such as a mission to Mars. To assess the potential of our Astropharmacy to meet the demand for pharmaceutical countermeasures, we compiled a database of biologics that could be produced by microbes and may be needed to mitigate the health risks during long-duration missions to the Moon and Mars. We focused on peptide and protein drugs as they can commonly be produced by microbes. This is the first formulary of peptide-based medications for deep-space missions and can help define the current landscape and opportunities to use this unique class of drugs in treating spaceflight-induced medical conditions. With this, we demonstrate that Astropharmacy has the potential to become a valuable resource for flight surgeons and for the medical community on Earth, particularly for those providing care in resource-limited environments.

2 Materials and methods

The methodology to compile the database was divided into six stages (Figure 2), as discussed in detail in the following sections.

2.1 Stage 1: establishing the database from historical ISS usage

The database began as a list of the pharmaceuticals named in the International Space Station Integrated Medical Group (IMG) Medical Checklist (Medical Operations Directorate and Operations Division, 2001). This comprehensive, 1000-page medical manual was published by NASA's Mission Operations Directorate in 2001 and was developed in collaboration with international partners to reflect the standards adhered to by all

astronauts on the ISS. This medical checklist defines protocols for astronaut health management to be used by the flight surgeon on Earth, who advises the crew in real-time, the crew medical officer designated to handle medical issues onboard, and all crew members (Yule et al., 2023).

2.2 Stage 2: addition of peptide drugs useful for long-duration missions

We reviewed the literature on peptide-based drugs that have been used on Earth to treat conditions that may also arise on long-duration deep-space missions. This literature was found primarily using PubMed and Google Scholar, with keywords such as “pharmaceutical countermeasures,” “microgravity,” “radioprotection,” and “space medicine.” Further research was then performed to determine which of these drug candidates were peptide-based and, therefore, would be added to the database. Considering the rapidly growing number of peptide-based drugs, we also consulted general lists of recently approved (or nearly approved) peptide-based drugs. These data were found in the published literature (Wang et al., 2022; Rossino et al., 2023). We reviewed the peptide-based drugs from these lists to identify those that are used on Earth to treat medical conditions that are also anticipated during deep-space missions. This included indications related to the treatment of radiation exposure, osteoporosis, neutropenias, pain, sleep difficulties, and other conditions. Once peptide-based drugs of interest were identified, the following information was compiled for each drug: common/brand names, function, drug class, FDA approval status and indication, shelf life and storage conditions, interest to NASA, amino acid length, and a review of prior transgenic production.

2.3 Stage 3: FDA approval status

FDA approval status was documented to indicate the drug development stage and whether the drug has demonstrated safety

and efficacy on Earth. Although drug effectiveness may be impacted in space, the U.S. crew onboard the ISS currently use commercial, FDA-approved medications. The “Drugs@FDA: FDA-Approved Drugs” database was used to determine a drug’s current approval status, while FDA drug labels available online were particularly useful for gathering information regarding drug storage and shelf life. For non-FDA-approved drugs, clinical trial history was searched on [ClinicalTrials.gov](https://clinicaltrials.gov), a database from the National Institute of Health. In some cases, information on shelf life and storage was instead obtained from the manufacturer’s website.

2.4 Stages 4–5: indications for use, storage conditions, and shelf life

Pharmaceutical company websites often contain information regarding drug indications, shelf life, and storage. Further basic information for each drug was collected from PubChem, a free and public database of chemical information maintained by the U.S. National Institute of Health, National Center for Biotechnology Information, and the U.K.’s electronic medicines compendium (eMC), an online database of regulated medicine maintained by Datapharm Ltd., an independent U.K. company, with oversight from the Medicines and Healthcare Products Regulatory Agency and the European Medicines Agency. PubChem offers chemical structures, physical and chemical properties, biological activities, toxicity information, pharmacological data, health and safety information, associations with peer-reviewed publications, and drug mechanisms. eMC is updated monthly by hundreds of pharmaceutical companies to document product characteristics and patient information. In some instances, we consulted other health departments outside the U.S. or the U.K. for additional information. For example, we referenced The Drug and Health Product Register, an online medicinal database maintained by Health Canada, and the Therapeutic Goods Administration from the Australian Government’s Department of Health and Ageing. DrugBank, a medicinal database maintained by the University of Alberta ([Wishart et al., 2006](#)), was also used to find amino acid sequence information.

2.5 Stage 6: transgenic production from PubMed and Google Scholar

To determine whether a peptide drug has been produced recombinantly (as opposed to chemical synthesis or isolation from an animal), we search the published literature on PubMed, Google Scholar, and Google Patents. Keywords such as the drug name followed by “recombinant production,” “transgenic production,” “production in *Bacillus subtilis*,” and “production in *Escherichia coli*” were used. Once an article or patent was identified, it was reviewed to determine which host expression system was used for peptide production. We particularly focused on microbial or yeast expression systems as they are best suited for spaceflight and, thus, have better utility for the Astropharmacy platform.

2.6 Data analysis and figure preparation

Data analysis and figure preparation were conducted using GraphPad Prism 10.1.2 (Dotmatics). The database was compiled in the web-based spreadsheet application Google Sheets (Google), which allowed for collaboration across the geographically distributed team, version control, and integration with other Google Workspace tools or file formats.

3 Results and discussion

3.1 Insights from current medications onboard the International Space Station

We reviewed 92 medications currently in use on the ISS to identify medications that could be produced recombinantly by microbes and, thus, could be investigated for production in the Astropharmacy platform. This platform would be ideal for the production of peptide-based drugs with a short shelf life or that require refrigeration or freezing for storage. Given the relatively short mission durations and non-medical countermeasures (e.g., diet and exercise), astronauts on the ISS generally do not have to rely on pharmaceutical countermeasures to mitigate the health risks associated with spaceflight. Thus, the current supply of medications largely serves as a “first aid kit” to treat common ailments (e.g., infections, colds, nausea, diarrhea, sleep issues, headaches, muscle soreness, and motion sickness) or medical emergencies (e.g., severe allergic reactions and heart issues). These common ailments can be treated with non-peptide drugs that generally have a longer shelf life and may be stored at room temperature. We found that only 1 out of 92 catalogued drugs, the bacitracin antibiotic, was peptide-based and could, therefore, potentially be produced on-demand in the Astropharmacy. It is naturally produced by *Bacillus licheniformis* and *B. subtilis*, although engineered biosynthesis of bacitracin has also been demonstrated in *B. subtilis* ([Eppelmann et al., 2001](#)). Additionally, the antifungal cream clotrimazole and the antiviral agent oseltamivir have peptide-based alternatives that may be good candidates for on-demand production using the Astropharmacy platform.

3.2 A database of peptide-based medications to mitigate the health risks in deep space

During deep-space missions, astronauts increasingly rely on pharmaceutical intervention to mitigate the health risks associated with the hazards of spaceflight. On-demand production of pharmaceuticals can mitigate the challenges such as limited shelf life, storage and up-mass constraints, and the inability to restock. We compiled a database of peptide-based medications used on Earth to treat conditions that may also be induced by spaceflight and could be investigated for production in the Astropharmacy. We identified 30 peptide drugs that are used on Earth but are not currently in use by NASA or other space agencies and organized them into eight categories based on their function ([Table 1](#)). The identified peptide drugs were compiled into a

TABLE 1 Categorical breakdown of identified peptide drugs with indications relevant to deep-space missions.

Drug category	Peptide drugs	% of database
Hematological	Tissue plasminogen activator, streptokinase, activated factor VII, granulocyte colony-stimulating factor, granulocyte-macrophage colony-stimulating factor, erythropoietin, romiplostim, hirudin, CBLB502, and CBLB612	33%
Musculoskeletal	Teriparatide, abaloparatide, salmon calcitonin, and amylin	13.3%
Cardiopulmonary	Nesiritide, angiotensin II, and vasopressin	10%
Metabolic	Insulin, glucagon, and dasiglucagon	10%
Antimicrobial	Daptomycin, PL-18, and teicoplanin	10%
Gastrointestinal	Linaclotide, interleukin-11, and TP508	10%
Analgesic	Leptucin and ziconotide	6.7%
Central nervous system	Delta sleep-inducing peptide and interleukin-18-binding protein	6.7%

database containing information on common/brand names, function, drug class, FDA approval status and indication, shelf life and storage conditions, interest to NASA, amino acid length, and a review of prior transgenic production (Table 2).

As summarized in Table 1, hematological drugs that generally treat conditions related to the blood, bone marrow, or the body's hemostatic mechanisms constitute 33% of the database. The identified hematological peptides can help mitigate the risks of radiation-induced blood conditions, such as neutropenia, blood clots, and sudden bleeding episodes during deep space missions. Peptides in the musculoskeletal category represent ~13% of the database, with indications for mitigating microgravity-induced bone loss. Cardiopulmonary, metabolic, antimicrobial, and gastrointestinal drug categories each make up 10% of the database. Finally, peptides in the analgesic and central nervous system categories constitute ~7% of the database each. Overall, the drug categories represented in the database align with the major risks to astronauts' health during deep-space missions.

The majority (21/30) of the identified peptides in Table 2 are <100 amino acids long, with only four peptides exceeding 200 amino acids and all being <550 amino acids long (Figure 3A). It is important to note that their small size does not necessarily correlate with ease of recombinant expression. For example, small peptides can be prone to proteolytic degradation when expressed in microbial host systems (Li, 2011). Nevertheless, the majority (24/30) of the identified peptides have been recombinantly produced (Figure 3B). Our review of recombinant production focused on microbes, such as *Escherichia coli* or *Bacillus subtilis*, and yeast as host organisms as these systems are better suited for on-demand production in deep space due to a simpler process (compared to mammalian hosts) and the ability to transport some of these host organisms as durable spores. One limitation of microbial systems is their limited ability for post-translational modifications. Although human peptides and proteins commonly contain post-translational modifications, these are not always necessary for bioactivity. A remarkable example of this is filgrastim, an FDA-approved recombinant human granulocyte colony-stimulating factor (G-CSF) that was successfully produced in *E. coli* without glycosylation (Welte et al., 1996). Additionally, progress has been made to engineer prokaryotic glycosylation pathways into microbes (Valderrama-Rincon et al., 2012; Palma et al., 2024). Overall, the examples of recombinant production in microbes or yeast that we

identified for the majority of the peptides in the database demonstrate the potential utility of a microbial-based system for on-demand pharmaceutical peptide production in deep space.

We demonstrated the expression of two peptides identified in Table 2, teriparatide and G-CSF, in *B. subtilis* (Vallota-Eastman et al., 2023). Our initial analysis focused on the need for G-CSF to treat severe neutropenia (low white blood count), a potentially severe medical condition that can result from a solar particle event (SPE) (Carnell, 2019). SPEs are unpredictable and can deliver a dose of radiation that is sufficient to cause severe neutropenia. To restore white blood cell count and prevent a possibly life-threatening infection, G-CSF would need to be administered within the first 3 days. Carnell (2019) highlighted that FDA-approved drugs for the treatment of neutropenia on Earth, such as G-CSF, could be life saving for the crew in deep space and, thus, should be considered for future medical toolkits.

Although we could not identify the recommended storage conditions and shelf life for five peptides that are in pre-clinical or early clinical trial stages, the majority of the peptides (17/30) require refrigeration or freezing for long-term storage (Figure 3C). Under recommended storage conditions, the average shelf life of the identified peptides (except the five where shelf-life information was not available) is ~30 months, which is the expected duration of a long-term mission on the Martian surface (Figure 4A). However, refrigeration and freezing capabilities are likely to be limited during deep-space travel due to up-mass, volume, and power constraints. When these drugs cannot be stored refrigerated or frozen as recommended, the shelf life decreases dramatically, often from years to mere weeks (Table 3). The shelf life may also be further reduced due to the spaceflight environment (e.g., radiation) and if the drugs are removed from their original packaging. Additionally, the shelf life can be greatly reduced once a drug is opened or reconstituted, even when continued to be stored under recommended conditions.

When shelf life and recommended storage conditions are considered together, peptide drugs that expire before the planned return to Earth and require refrigeration or freezing would be impractical for deep-space missions, even if they are proven safe and effective in space and would, therefore, be ideal candidates for on-demand production using the Astropharmacy platform. Twelve of the peptides identified as potentially suitable for space meet these shelf life and storage criteria, including four hematological, three

TABLE 2 The database of 30 peptide-based drugs with pharmaceutical names, function, drug class, FDA approval status and indication, shelf life and storage, NASA interest, amino acid length, and a review of prior transgenic production.

Pharmaceutical name	Common/ Brand name	Function/ Mechanism	Drug class	FDA approval (as of 07/2025) and indications	Shelf life and storage conditions	NASA interest	# Amino acid	Prior transgenic production	Source
Tissue plasminogen activator (tPA)	Alteplase, reteplase, and tenecteplase	Dissolution of blood clots	Thrombolytic agent	Approved for stroke and ST elevation myocardial infarction (STEMI)	12 months at −20 °C or 3 weeks at room temperature	Emergency dissolution of blood clots	355–527	Yes [e.g., <i>E. coli</i> , <i>B. subtilis</i> , <i>P. pastoris</i> , and Chinese hamster ovary (CHO) cells]	Lin-Fa et al., 1989; Khan et al., 1995; Qiu et al., 1998; Mohammadi et al., 2019; Jilani and Siddiqui, 2025; Acris Antibodies, 2025
Streptokinase	Kabikinase and Streptase	Originally approved for STEMI, NSTEMI, unstable anginas, pulmonary embolism, deep-vein thrombosis, arterial thrombosis or embolism, and occlusion of arteriovenous cannulae	Thrombolytic agent	No longer commercially available	36 months unopened at 20 °C–25 °C	Emergency dissolution of blood clots	440	Yes (e.g., <i>E.coli</i> and <i>S. pyogenes</i>)	Mundada et al., 2003; Assiri et al., 2014; Aghaeepoor et al., 2019; eMC database
Activated factor VII	Coagulation factor VIIa (recombinant)–jncw or SEVENFACT NOVOSEVEN	Blood clotting factor for hemorrhage	Clotting factor	Approved for bleeding episodes and prevention in surgical procedures on hemophiliacs	36 months in desiccated form and stored below 25 °C	Sudden bleeding episodes	406	Yes [e.g., <i>Leishmania</i> , <i>Spodoptera frugiperda</i> , and baby hamster kidney (BHK) cells]	Goudemand, 1998; Wishart et al., 2006; Masroori et al., 2010; Mirzaahmadi et al., 2011; LFB, 2024; European Medicines Agency database
Granulocyte colony-stimulating factor (G-CSF)	Lenograstim, filgrastim, Neupogen, and Neulasta	Stimulates the production of granulocytes	Colony-stimulating factor	Approved for neutropenia	12 months when stored at −20 °C to −80 °C	Treatment of radiation-induced neutropenia, particularly in the lungs	175	Yes (e.g., <i>E. coli</i> , <i>B. subtilis</i> , and <i>S. cerevisiae</i>)	Bae et al., 1999; Bashir et al., 2015; Trinh et al., 2021; Skokowa et al., 2022; Cell Guidance Systems, 2025
Granulocyte–macrophage colony-stimulating factor (GM-CSF)	Sargramostim (Leukine) Molgramostim (Molgradex)	Stimulates proliferation and differentiation of hematopoietic progenitor cells	Colony-stimulating factor	Sargramostim approved for post-chemotherapy, post-blood or post-bone-marrow transplantation and for acute radiation syndrome	12 months stored at −20 °C to −80 °C or 1 month at 2 °C–8 °C	Treatment of radiation-induced lung injury	127	Yes (e.g., <i>E. coli</i> and <i>P. pastoris</i>)	Gajewska et al., 2018; U.S. Food and Drug Administration, 2018; Pykhtina et al., 2022; Biotechnie R&D Systems, 2024

(Continued on following page)

TABLE 2 (Continued) The database of 30 peptide-based drugs with pharmaceutical names, function, drug class, FDA approval status and indication, shelf life and storage, NASA interest, amino acid length, and a review of prior transgenic production.

Pharmaceutical name	Common/ Brand name	Function/ Mechanism	Drug class	FDA approval (as of 07/2025) and indications	Shelf life and storage conditions	NASA interest	# Amino acid	Prior transgenic production	Source
Erythropoietin	Epoetin alfa, Epogen, and Procrit	Stimulates production of erythrocytes and erythrocyte precursors in the bone marrow	Hematopoietic growth factor	Approved for treatment of anemia	18 months with refrigeration (2 °C–8 °C)	Treatment of acute radiation syndrome	165	Yes (e.g., CHO cells, <i>E. coli</i> , and <i>C. reinhardtii</i>)	Eichler-Stahlberg et al., 2009; Grunina et al., 2017; Susantad et al., 2021; eMC database
Romiplostim	Nplate	Increases platelet and megakaryocyte production	Thrombopoietin receptor agonist/ colony-stimulating factor	Approved for thrombocytopenia	36 months at 2 °C–8 °C, protected from light 30 days at RT 24 h once reconstituted	Treatment of radiation-induced thrombocytopenia	528 (2 identical single-chain subunits of 269	Yes (e.g., <i>E. coli</i>)	Amgen Inc, 2008; Australian Government, 2008; Yang, 2015; PubChem database
Hirudin	Lepirudin and Refludan	Inhibits coagulant activity of thrombin	Thrombin inhibitor	Discontinued. Was originally FDA-approved as an anticoagulation in patients with heparin-associated thrombocytopenia (HIT) and associated thromboembolic disease	36 months unopened at 20 °C–25 °C. Do not freeze or do not store over 25 °C. Use immediately after reconstitution	Sudden bleeding episodes	65	Yes (e.g., <i>E. coli</i> , <i>S. cerevisiae</i> , and <i>P. pastoris</i>)	Zhang and Lan, 2018; PubChem database; European Medicines Agency database
CBLB502	Entolimod	Reduces radiation-induced apoptosis and accelerates the regeneration of progenitors in radiation-damaged tissues	Toll-like receptor (TLR) 5 agonist	Not FDA-approved, phase II clinical trials	Not available	Use for acute radiation syndrome. Protect hematopoietic and GI systems and stimulate regeneration	329	Yes (e.g., <i>E. coli</i>)	Ye et al., 2018; European Medicines Agency, 2018; Singh and Seed, 2021; NCT02715882
CBLB612	N/A	Inhibits growth hormone receptor, potentially inhibiting growth and proliferation of cancer cells	TLR 2 agonist	Not FDA-approved, phase II clinical trials	Not available	Use for acute radiation syndrome	9	No	Yang et al., 2021; NCT02778763
Teriparatide	Forteo	Stimulates bone formation through the stimulation of osteoblastic activity	Parathyroid hormone analog	Yes, for treatment of osteoporosis	24 months unopened at 4 °C–8 °C 28 days once opened at 4 °C–8 °C	Counteract loss of bone density due to microgravity	34	Yes (e.g., <i>E. coli</i> and <i>B. subtilis</i>)	Eli Lilly, 2008; Vallota-Eastman et al., 2023; Medicines Management Programme, 2024; PubChem database

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TABLE 2 (Continued) The database of 30 peptide-based drugs with pharmaceutical names, function, drug class, FDA approval status and indication, shelf life and storage, NASA interest, amino acid length, and a review of prior transgenic production.

Pharmaceutical name	Common/ Brand name	Function/ Mechanism	Drug class	FDA approval (as of 07/2025) and indications	Shelf life and storage conditions	NASA interest	# Amino acid	Prior transgenic production	Source
Abaloparatide	Tymlos	Stimulates osteoblast activity, increasing bone density	Parathyroid hormone analog	Yes, for treatment of osteoporosis	36 months at 2 °C–8 °C	Counteract loss of bone density due to microgravity	34	Yes (e.g., <i>E. coli</i>)	Wang et al., 2014; PubChem database; European Medicines Agency database
Salmon calcitonin	Miacalcic calcitonin	Decreases osteoclast activity and thereby bone resorption. Stimulates calcium salt deposition in bone	Calcitonin	Yes, for postmenopausal osteoporosis, hypercalcemia, and Paget's disease	60 months unopened under refrigeration (2 °C–8 °C)	Counteract loss of bone density due to microgravity	32	Yes (e.g., <i>E. coli</i>)	Hong et al., 2000; Novartis, 2014; McLaughlin et al., 2025
Amylin	IAPP	Decreases bone resorption and increases osteoblast formation	Hormone	No. Clinical studies in progress	Not available	Counteract loss of bone density due to microgravity	37	Yes (e.g., <i>E. coli</i>)	Krampert et al., 2000; Noh et al., 2020; NCT06186063
Nesiritide	Natrecor	Regulates cardiovascular homeostasis through vasoconstriction	Natriuretic peptide	Discontinued, originally FDA-approved for acutely decompensated heart failure	48 months at 15 °C–30 °C Reconstituted solution must be used within 24 h and stored between 2 °C and 25 °C	Use for emergency cases of heart failure	32	Yes (e.g., <i>E. coli</i>)	Scios Inc, 2001; Rao et al., 2011; Health Canada database
Angiotensin II	Giapreza	Increases blood pressure through vasoconstriction, fluid reabsorption, and release of aldosterone	Vasopressor	Yes, for low blood pressure with septic or distributive shock	36 months for unopened vial at 2 °C–8 °C One diluted, 24 h at room temperature or refrigeration (2 °C–8 °C)	Use for hypotension due to head trauma or other causes including dehydration	8	Yes, for the precursor ovine ANG (e.g., <i>E. coli</i> and <i>P. pastoris</i>)	Yamashita et al., 2016; Palanikuma et al., 2019; European Medicines Agency database
Vasopressin (ADH)	Vasopstrict	Increases blood pressure through vasoconstriction and fluid reabsorption	Vasopressor	Yes, to increase blood pressure in patients with vasodilatory shock	12 months at 20 °C–25 °C, 24 months at 2 °C–8 °C. Once diluted, 18 h at room temperature, or 24 h under refrigeration	Use for hypotension due to head trauma or other causes including dehydration	9	Yes (e.g., cell-free wheat germ system)	American Regent Inc, 2020; Sterns, 2020; Endo Inc, 2025; Novus Biologicals, 2025

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TABLE 2 (Continued) The database of 30 peptide-based drugs with pharmaceutical names, function, drug class, FDA approval status and indication, shelf life and storage, NASA interest, amino acid length, and a review of prior transgenic production.

Pharmaceutical name	Common/ Brand name	Function/ Mechanism	Drug class	FDA approval (as of 07/2025) and indications	Shelf life and storage conditions	NASA interest	# Amino acid	Prior transgenic production	Source
Linaclotide	Linzess	Increases secretion of solutes into intestinal lumen, resulting in increased secretion and softer stool	Guanylate cyclase C-agonists	Yes, for constipation and irritable bowel syndrome	24–36 months unopened at 20 °C–25 °C 18 weeks after opening	Change in diet, environment, and stress may cause constipation. Previous astronauts have reported constipation	14	Yes	Thomas and Allmond, 2013; AbbView Inc, 2024; Kalola et al., 2025; European Medicines Agency database; patent # WO2016038497A1
Interleukin-11 (IL-11)	Neumega (oprelvekin)	Stimulates the production of hematopoietic stem cells, resulting in increased platelet production	Thrombopoietic growth factor	Discontinued. Originally FDA-approved for prevention of severe thrombocytopenia and reduction in the need for platelet transfusions following myelosuppressive chemotherapy	24 months at 2 °C–8 °C	Radiation protection, particularly for the gut	180 (recombinant version)	Yes (e.g., <i>E. coli</i> , <i>P. pastoris</i> , and <i>S. cerevisiae</i>)	Lipiäinen et al., 2015; BPS Bioscience, 2025; patent # US20210317500A1; Drugs.com database
Rusalatide acetate (TP508)	Chrysalin	Activates radioresistant stem cells and increases the stemness potential of crypts to maintain and restore intestinal integrity	Regenerative peptide	No, phase II clinical trials for radiation-induced gastrointestinal damage	72 months at –20 °C	Mitigate radiation-induced intestinal and colonic toxicity post-radiation exposure	23	No	Ryaby et al., 2006; Kantara et al., 2015; Chrysalis Biotherapeutics, 2024; NCT00131482; patent # WO2020210087A1
Insulin	Lantus, Humalog, and Tradjenta	Transports glucose and potassium from the blood into cells	Hormone	Yes, for glycemic control in patients with diabetes	12 months with refrigeration (4 °C–6 °C)	Treatment of hyperkalemia, which may arise due to conditions such as renal failure	51	Yes (e.g., <i>E. coli</i> , <i>S. cerevisiae</i> , and <i>B. subtilis</i>)	Feng et al., 2013; Baeshen et al., 2014; U.S. Food and Drug Administration, 2017; Donnor and Sarkar, 2023
Glucagon	GlucaGen	Increases blood glucose by promoting glycogenolysis and gluconeogenesis	Glucose-elevating agent and hormone	Yes, for hypoglycemia in diabetic patients	24 months at room temperature	Necessary precaution if insulin is to be carried	29	Yes (e.g., <i>E. coli</i>)	Ishizaki et al., 1992; Novo Nordisk Medical, 2025; PubChem database
Dasiglucagon	Zegalogue	Increases blood glucose by promoting glycogenolysis and gluconeogenesis	Glucose-elevating agent and hormone	Yes, for hypoglycemia in diabetic patients	12 months at room temperature	Necessary precaution if insulin is to be carried	29	Information regarding production unavailable	Rechon Life Science AB, 2021; Al Shaer et al., 2022; PubChem database

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TABLE 2 (Continued) The database of 30 peptide-based drugs with pharmaceutical names, function, drug class, FDA approval status and indication, shelf life and storage, NASA interest, amino acid length, and a review of prior transgenic production.

Pharmaceutical name	Common/ Brand name	Function/ Mechanism	Drug class	FDA approval (as of 07/2025) and indications	Shelf life and storage conditions	NASA interest	# Amino acid	Prior transgenic production	Source
Delta sleep-inducing peptide	DSIP and Emideltide	Increases the production of GABA, slowing brain activity, inducing relaxation, and promoting deeper sleep	Neuromodulator	Not FDA-approved	12 months at –80 °C 1 month at 2 °C–8 °C	Use to promote sleep—on orbit, sleep problems are common and may detriment health and performance	9	Yes (e.g., <i>E. coli</i> and <i>P. pastoris</i>)	Oh et al., 2003; Mu et al., 2024; Abbexa, 2025
IL-18 binding proteins (IL-18 BP)	Tadekinig alfa	Inhibits IL-18, the key protein in radiation-induced cell and tissue damage and dysfunction	Immunoglobulin	Not FDA-approved. Phase III trials	6 months below –70 °C Post-reconstitution 1 week at 4 °C or 3 months at –20 °C	Use for radioprotection	164	Yes (e.g., <i>E. coli</i>)	Li et al., 2020; Guo et al., 2023; GenScript 2025; NCT03113760
Leptucin	Leptucin	May act on supraspinal sites and spinal opioid receptors (being investigated)	Analgesic	Not FDA-approved. In the pre-clinical stage	Not available	Use to treat acute pain	76	No	Bagheri-Ziari et al. (2021)
Ziconotide	Prialt	Selectively blocks N-type calcium channels on nociceptive afferent nerves	Analgesic	Yes, for severe chronic pain management	48 months at 2–8 °C. 24 h once reconstituted at 2 °C–8 °C	Treatment of chronic pain	25	No	eMC database; PubChem database
Daptomycin	Cubicin	Inhibits bacterial cell wall integrity, along with DNA and RNA synthesis	Antibiotic	Yes, for Gram-positive bacterial skin infections	36 months at 2 °C–8 °C. 12–48 h once reconstituted	Antimicrobials may be useful as astronauts may have a weakened immune system in space	13	Yes (e.g., <i>S. lividans</i>)	Penn et al., 2006; Tzotzos, 2016; Heidary et al., 2018; eMC database
PL-18	N/A	May inhibit quorum sensing, disrupt metabolic processes, and prevent biofilm formation (being investigated)	Antimicrobial and antifungal	No. Phase I trials	Not available	Use for vaginal tract infections; hygiene challenges of spaceflight may increase incidence of such infections	15	No	Butler et al., 2023; Rossino et al., 2023; Schlichter Kadosh et al., 2024; NCT05340790; patent # EP2735570B1
Teicoplanin	Targocid	Prevents formation of peptidoglycan, inhibiting bacterial cell wall synthesis	Antibiotic	No. Approved in Europe for Gram-positive bacterial skin infections	36 months when stored at 20 °C–25 °C in powder form 24 h once reconstituted and under refrigeration	Antimicrobials may be useful as astronauts may have a weakened immune system in space	7 (within the core structure)	No	Economou et al., 2013; Vimberg, 2021; Sanofi, 2019

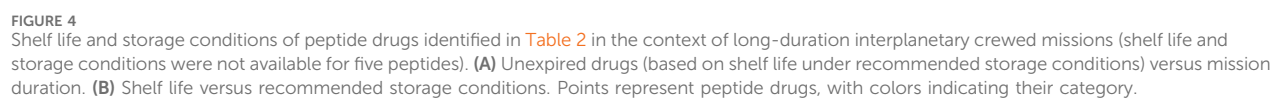
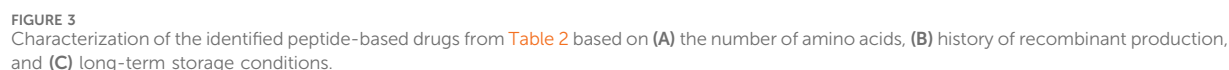


TABLE 3 Changes in shelf life when drugs are moved from sub-8 °C storage to room temperature (20 °C–25 °C). This table included drugs listed in Table 2 for which cold storage is recommended but for which information on room-temperature shelf life is also available.

Small-peptide drugs	Shelf life when frozen or refrigerated (as indicated)	Shelf life at room temperature (unless specified)	Decrease in shelf life
Tissue plasminogen activator	12 months	21 days	94.2%
Erythropoietin	18 months	3 days	99.5%
Insulin	12 months	28 days	92.3%
Abaloparatide	36 months	30 days	97.3%
Vasopressin	24 months	365 days	50%
Ziconotide	48 months	60 days (at 37 °C)	95.9%
Romiplostim	36 months	30 days	97.3%

metabolic, two central nervous system, and one from each of the musculoskeletal, cardiopulmonary, and gastrointestinal drug categories (Figure 4B). As humans undertake long-duration interplanetary crewed missions, on-demand, on-site production may become the only option for providing pharmaceuticals to mitigate health risks.

3.3 Database limitations, utility, and future direction

It should be noted that the presented database is not an exhaustive list of useful peptide drugs for deep space, but rather a foundational list that is designed to be dynamic and built upon. To identify additional pharmaceutical peptides of interest for the database, a systematic review of all approved peptide drugs, followed by cross-referencing their indications with the identified human health risks in deep space (e.g., as identified by the Human Research Program), would be beneficial. One of the challenges encountered in generating the database is that some peptide drugs have multiple commercial alternatives, which may vary in shelf lives, storage conditions, and amino acid sequences. Additionally, we were unable to locate storage and shelf-life information for some of the identified peptides.

Although the database currently only includes peptide-based drugs, the addition of recombinant monoclonal antibodies (mAbs) could be explored in the future. The number of mAb treatments has sharply increased in recent years (Mahmuda et al., 2017), and mAbs have the potential to treat several conditions, from infectious diseases to osteoporosis. Furthermore, mAbs provide useful diagnostic capabilities by detecting biomarkers for various diseases or identifying the presence of events such as a heart attack (Quinteros et al., 2017). Currently, recombinant mAbs are mainly produced in mammalian cells, which provide the proper folding and glycosylation patterns. However, antibody fragments, which do not require the same post-translational modifications as full antibodies, have been produced in *B. subtilis*. These antibody fragments can have the same binding capacity as full antibodies, allowing for their clinical use (Lakowitz et al., 2017). Although the clinical efficacy of antibody fragments and their production in bacterial host systems is still being studied, integrating mAbs or

antibody fragments into the Astropharmacy platform could open the door to a wider variety of therapeutics and diagnostic molecules available to astronauts. In addition to mAbs, the database and Astropharmacy capabilities can be expanded to include other “natural product” agents, for example, penicillin family antibiotics, antifungal agents, and some chemo agents.

Although Artificial Intelligence (AI) was not used in the production of this database, it could be a valuable tool in the future. AI could help comb through massive amounts of the literature and datasets to help identify drugs to add to the database, along with providing information such as drug stability and recombinant production capability. Importantly, AI models could be created to help predict the challenges of producing specific biologics in the database. A peptide production “difficulty estimate” could be generated from historical information from production trials, including details on peptide size, post-translational modifications, and integration of genetic and metabolic characteristics of the host expression system. Additionally, AI could be used to streamline the overall format, readability, and “look” of the database, optimizing it for users.

4 Conclusion

The medical kits onboard the ISS are composed of first-aid supplies and medications that target common ailments. Although these medications will still be needed on deep-space missions, there will also be a need for pharmaceutical countermeasures to address the unique health risks associated with deep-space travel. To our knowledge, this is the first study to compile a list of pharmaceutical peptides and proteins that have been used on Earth for the treatment of medical conditions that may also arise during deep-space missions. These peptides and proteins are commonly produced using recombinant organisms on Earth and thus have the potential to be produced on demand in deep space using platforms such as Astropharmacy. This could mitigate storage concerns and open the door for including peptide- and protein-based pharmaceuticals in the medicinal formulary for human deep-space missions. We aim to establish and demonstrate on-site, on-demand capabilities to produce the identified peptides using the Astropharmacy platform, whereas the compiled database can guide

future research into the safety and efficacy of these peptides in space and serve as a useful resource for flight surgeons and those developing mission architectures for future human exploration of deep space.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material; further inquiries can be directed to the corresponding author.

Author contributions

RB: Conceptualization, Methodology, Investigation, Writing – original draft. DO: Conceptualization, Methodology, Investigation, Writing – original draft. AK: Visualization, Writing – original draft. JS: Conceptualization, Visualization, Writing – review and editing. CB: Conceptualization, Investigation, Writing – review and editing. PW: Conceptualization, Methodology, Supervision, Writing – review and editing. DL: Conceptualization, Methodology, Supervision, Writing – review and editing. LR: Conceptualization, Methodology, Supervision, Writing – review and editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

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